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REVIEW ARTICLE

Advancing Anaerobic Digestion of Biodiesel Byproducts: A Comprehensive Review

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Abstract
The energy crisis, climate change, and insufficient waste management practices are compelling factors driving research into sustainable waste-to-resource technologies. Anaerobic digestion, aiming to recover energy and nutrients from organic waste, aligns with the circular economy's principles. This paper provides a comprehensive overview of utilizing biodiesel byproducts for biogas production, exploring techniques for enhancing biogas yield and addressing associated challenges. Assessing the potential of biodiesel byproducts highlights their environmental sustainability and economic viability for biogas production. Non-edible seed cake, rich in nutrients, shows promise for significant biogas yield. Additionally, crude glycerol, easily biodegradable, is identified as a promising co-digester, aiding in digesting recalcitrant substrates. Empirical data reveals remarkable methane yield boosts, ranging from 14 to 226% when co-digesting with crude glycerol. Moreover, the resulting digestate enhances soil fertility, promoting healthier plant growth and productivity. Challenges in anaerobic digestion, such as substrate C/N ratio imbalance and recalcitrance, necessitate strategies like substrate pretreatment and co-digestion with compatible materials to optimize biogas yield. Furthermore, advancements in anaerobic digestion technologies are crucial for effectively converting biodiesel wastes into biogas. Additionally, interdisciplinary investigations, including techno-economic analysis, lifecycle assessment, and sensitivity analysis, are vital to enhance and validate the feasibility of anaerobic digestion for biodiesel byproducts. This review serves as a valuable resource for future utilization of biodiesel byproducts for biogas production.

Keywords Additives · Bioenergy · Circular bioeconomy · Crude glycerol · Digestate · Seed cake · Waste-to-resource

Introduction

The world's energy consumption is mainly fulfilled by conventional fuels, with the transportation sector accounting for about 29% of world energy demand [1] and responsible for more than 60% of fossil fuel usage [2]. The rising energy demand and associated greenhouse gas emissions from fossil fuels are prompting researchers and policymakers to focus on searching for renewable energy sources. On the other hand, the issue of global waste generation has escalated significantly due to population growth, urbanization, economic development, and increased industrial activities. For example, in 2016, a total of 2.02 billion tons of waste was generated, a figure projected to climb to 2.59 billion tons by 2030 and 3.4 billion tons by 2050 [3]. Agricultural wastes have contributed to the increase in waste generation worldwide, driven by factors such as rising food production and consumption and rising use of raw materials in industrial processes. According to the 2021 reports, quantitative studies of agricultural waste revealed that, on average, 1.3 billion tons of agricultural waste are produced globally each year [4]. However, the management of waste particularly in developing countries often proves inadequate, leading to practices such as dumping in an open environment with associated

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environmental impacts. Addressing these challenges requires the development of cost-effective energy conversion technologies to utilize potential biomass for biobased renewable energy generation.

Renewable energy sources encompass wind, hydro, solar, geothermal, and biomass, particularly derived from organic waste materials. Biofuels, which can serve as alternatives to conventional fossil fuels, are produced from biological sources like plant, algal, and animal biomass, as well as from industrial and agricultural waste. Currently, biofuels are attracting considerable interest for their wide range of biomass sources and their potential to contribute to net-zero emissions. Nevertheless, global energy demand from biofuels and other renewables remains unmet, a gap that is especially evident in developing countries. These regions possess substantial biomass resources for biofuel production but continue to rely heavily on fossil fuels and traditional biomass. Biodiesel, among biofuels, is notable for being biodegradable and nontoxic, making it a valuable diesel alternative that contributes to lowering carbon emissions, particularly within transportation. Additionally, incorporating diverse biomass resources, including industrial byproducts, is crucial to bolstering the sustainability of bioenergy production.

The conversion of biodiesel byproducts into other forms of biofuel is one of the effective and sustainable waste management strategies to come with resource recovery, environmental sanitation, and pollution prevention. In general, the sustainable waste-to-energy conversion techniques of industrial wastes are classified into biochemical or thermochemical processes [3]. The biochemical pathway involves the use of microorganisms and enzymes to break down large molecules into intermediates in a way suitable to produce biofuels and other chemicals. In the environmental point of view, the biochemical energy conversion pathway is preferred over the thermochemical pathway [5]. This review is focused on the biochemical biomass waste valorization process. The biochemical energy conversion pathway includes anaerobic digestion, fermentation, and photobiological hydrogen production. Thus, anaerobic digestion is an exceptionally unique energy conversion technology which has the potential of producing biogas a zero net emission energy and nutrient-rich digestate.

Biogas is biofuel energy obtained from a wide range of biomass feedstocks, which are renewable and low-cost resources. Biogas energy is a fast-growing biofuel technology expected to increase its capacity four times by 2050 in Europe, which might take up to 40% of European gas demand [6]. Successful implementation of biogas technology resolves environmentally damaging and socially sensitive issues of fossil fuel dependence, greenhouse mitigation, stabilized product formation, energy recovery, and waste utilization into profitable solutions, producing energy and bio-fertilizers. Biogas technology also provides biomethane for transportation through upgrading technologies and compressing it like natural gas to power engines. Upgrading is the technique of biogas purification, or the removal of impurities from biogas to produce biofuel comparable to natural gas, which is known as biomethane.

Biogas technology mainly relies on the types of feedstocks used and the conversion technologies. Biofuel feedstocks are generally classified into first-, second-, third-, and fourth-generation feedstocks. The first-generation biofuels are derived from edible crops such as sugarcane for ethanol and oil crops for biodiesel. The critical problem with those energy sources is the competition with human food and nutrition, which leads to food insecurity [7]. These days, second-generation feedstocks (non-food crops or wastes from first-generation feedstocks), third-generation feedstocks (algal biomass), and fourth-generation feedstocks (genetically modified algae), are the potential candidates for biofuel production. Biogas is an exceptional type of biofuel due to its ability to be generated from recovered waste biomass resources and produce both energy and nutrient-rich fertilizer. Hence, many varieties of organic materials can be used as biogas feedstocks; however, highly biodegradable biomass materials are preferred as feedstocks. In general, the properties and composition of the feedstocks and substrates used in biogas production have a significant effect on energy efficiency. Senol [8] suggested that the important constituents of organic matter for the effective conversion of biomass into biogas are crude protein, fiber, cellulose, hemicellulose, sugar, and starch. Moreover, proteins and carbohydrates are characterized by a faster rate of decomposition compared to fat composition [9]. Additionally, the sustainability and availability of the feedstock in the vicinity of the biogas plant is another crucial factor in biogas technology.

The achievement of the Sustainable Development Goals (SDGs), including the creation of a clean environment, relies on implementing innovative waste management strategies that recover energy and nutrients simultaneously. This necessitates the exploration of potential new feedstocks and the adoption of advanced technologies for the efficient and sustainable conversion of biomass resources in a cost-effective manner. Although the availability of crude glycerol and seed cake from the biodiesel industry is increasing, impurities in crude glycerol and toxic compounds in non-edible seed cake present significant challenges to their optimal utilization. While purifying crude glycerol can be costly, biological waste treatment for biogas production offers a viable alternative. Research on the bioconversion of these wastes has shown technical feasibility and notable improvements in anaerobic digestion yields. However, using crude glycerol alone or in high concentrations during anaerobic digestion can inhibit methane-producing microbes, resulting in lower biogas and methane output. Thus, an effective solution to address these challenges and maximize the utilization of biodiesel byproducts remains to be developed.

The motivation of the present review is to explore the potential of utilizing industrial wastes, with a focus on biodiesel byproducts, for biogas generation through anaerobic digestion. Thus, the high nutrient content of seed cakes and the biodegradability of crude glycerol from the biodiesel industry have the potential to produce biogas through optimized anaerobic digestion. Current research has pointed to crude glycerol as a promising co-substrate for the anaerobic digestion of organic wastes. Nevertheless, there is no detailed established protocol regarding the maximum percentage of glycerol that can boost methane yield without overwhelming or impairing the digester. Moreover, seed cakes are nutrient-rich leftovers from oil extraction for biodiesel production, which needs to be utilized and is better co-digested with other carbon-rich substrates. Hence, it is necessary to explore alternatives for the effective use of the leftovers for energy production through the advancement of anaerobic digestion, by effectively reviewing previously published works. Co-digestion with other substrates dilutes inhibitors in crude glycerol, maintains buffering capacity, and enhances biogas yield by fostering synergism, supplying missing nutrients, and improving microbial diversity. Effective co-digestion requires substrates rich in nitrogen and micronutrients, consideration of crude glycerol composition, and reactor type. Optimizing operating conditions is essential to manage microbial tolerance to glycerol impurities. Innovative technologies, such as supplements, can further stabilize and enhance biodiesel byproduct bioconversion. The success of such additives depends on substrate type, digester configuration, microbial community, and operational parameters, requiring careful optimization of their composition and dosage.

Attention has been given to the approach of industrial waste valorization through anaerobic digestion in conjunction with the principles of the circular economy. This approach primarily focuses on recycling and reusing waste with the goal of minimizing waste generation to achieve zero waste. The main objective of this review is to assess the feasibility of utilizing biodiesel byproducts as potential resources for biogas production, advancing the anaerobic digestion of biodiesel byproducts, and expanding their anaerobic digestion by combining them with other available and low-cost substrates. To achieve this objective, this review explores (1) anaerobic digestion in general, with a particular focus on biodiesel byproducts; (2) techniques for maximizing energy conversion; (3) the energy potential and nutrient recovery of these byproducts; (4) the challenges and limitations of anaerobic digestion; and (5) recent advances in anaerobic digestion technologies.

This review, which examines relevant publications from 2010 onward, highlights the potential of biodiesel wastes as substrates or co-substrates for biogas production and discusses the status of substrate valorization. Specifically, it examines the use of wastes from the biodiesel industry, including crude glycerol and non-edible seed cake, as substrates for biogas production. The study evaluates their biogas production potential when combined with other substrates and addresses challenges such as the inhibitory properties of glycerol and the resistance of seed cake during anaerobic digestion. Furthermore, the study highlights the importance of incorporating economic models and adopting a circular bioeconomy approach, including life cycle assessments, to optimize byproduct valorization.

Anaerobic Digestion

Historical evidence shows that the implementation of biogas through aerobic digestion has been practiced for a long time. As evidence, biogas plants were built in the eighteenth century primarily in Asia and then transferred to Europe [10]. However, the actual anaerobic digestion technology was forwarded during World War II, and since then, many European countries have established many biogas plants. Anaerobic digestion technology is a type of biorefinery technology in which a mixture of biowastes is digested and converted into valuable products. The target of optimizing anaerobic digestion-based biorefinery is to maximize the efficient use of resources, minimize waste, and lastly maximize biogas yield. Additionally, sustainable biogas production systems include processes for treating waste, valorizing low-value products, and producing electricity, heat, and other good-value chemicals. The chemical energy released from the anaerobic digestion process as biogas is mainly composed of methane (55–65%), carbon dioxide (25–45%), small amount of water vapor, and other impurities. Anaerobic digestion is carried out through four stages: namely hydrolysis, acidogenesis, acetogenesis, and methanogenesis. The block diagram of the anaerobic digestion process modified by Rehman et al. [11] is depicted in Fig. 1.

After biomass pretreatment, the bacterium in the digester needs to access the energy potential of the organic matter through the breaking down of the polymeric material into smaller constituent parts. Hence, hydrolysis is the first step of anaerobic digestion, which breaks down macromolecules or polymers in the presence of certain bacteria to solubilize them in organic acids and alcohols formed by the action of microorganisms. In the acidogenesis stage, soluble molecules from the hydrolysis stage are converted into carbon dioxide (CO_2) and hydrogen (H_2) with the help of acidogenic bacteria. The important acid produced in the acidogenesis stage is acetic acid (CH_3COOH), a substrate for the methane-forming microorganisms [12]. The acetogenesis stage is the acetate-producing stage through the utilization of hydrogen by CH_4-producing

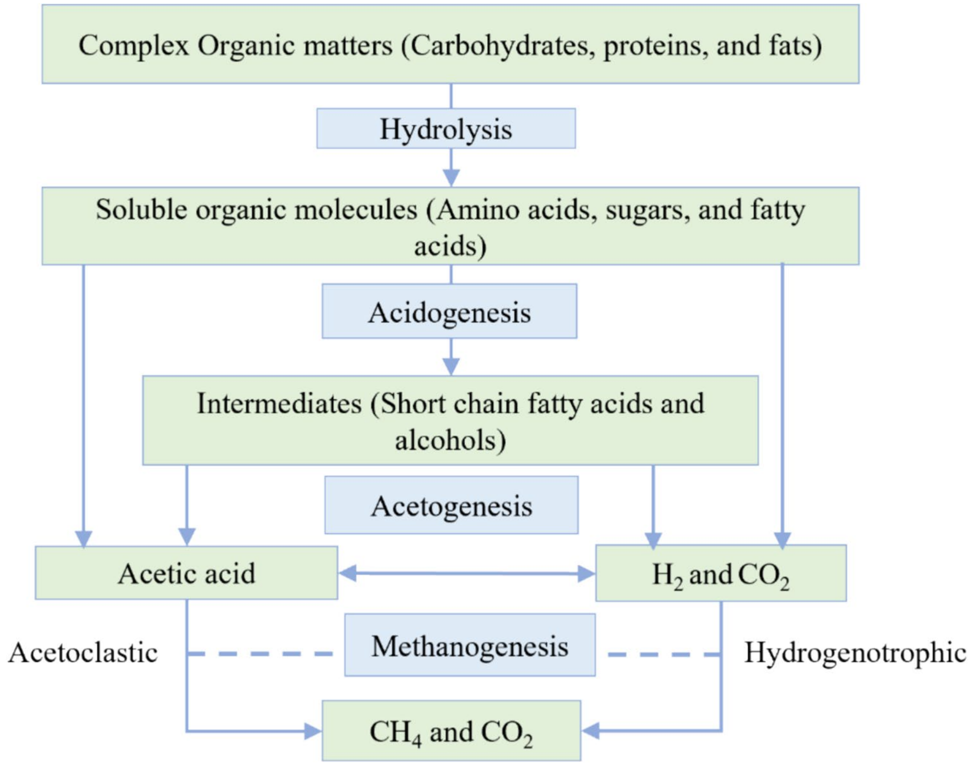

Fig. 1 Steps in anaerobic digestion process [11]

bacteria that can convert some of the bacteria into CH_4. At the methane formation stage, low molecular weight molecules (carbon dioxide and acetate) decompose further in the presence of methane-producing bacteria. In general, methane-producing bacteria can be grouped into two categories: (1) acidophilic CH_4 production is carried out by decarboxylation of acetate, and (2) hydrogenophilic CH_4 generation is achieved by reduction of H_2/CO_2 [13].

Moreover, the production of medium-chain volatile fatty acids in the anaerobic fermentation (acidogenesis and acetogenesis) stage of anaerobic digestion has attracted more attention. Hence, the volatile fatty acids produced during anaerobic digestion have the potential to produce value-added chemicals [14]. The most common volatile fatty acids that are intermediate in an anaerobic digestion process are acetic acid, propionic acid, and butyric acid. In the anaerobic digestion processes, organic acids are digested and mixed volatile fatty acids (acetic acid, propionic acid, and butyric acid) are produced. The application of the mixed volatile fatty acids includes the production of bioplastics, biofuels (bio-oil, bio-hydrogen, and bioethanol), and biological nutrient removal [15].

Environmental Benefit and Energetic Potential of Anaerobic Digestion

Anaerobic digestion, a biological waste management technology, provides a multifaceted solution to managing industrial wastes. As an eco-friendly technology, it repurposes diverse waste streams into biogas, volatile fatty acids, and nutrient-rich byproducts, thereby reducing the harmful effects of waste in the environment. These byproducts, known as digestate, are particularly significant as they serve as a sustainable alternative to conventional mineral fertilizers. By utilizing digestate, we can mitigate the financial and environmental impacts associated with mineral fertilizers. In addition, wastewater from agricultural wastes can be recycled and reused for sustainable agricultural production Moreover, the recycling and reuse of wastewater from agricultural wastes can further contribute to sustainable agricultural production [16].

Anaerobic digestion specifically for organic waste material has the advantage of reducing odors and pathogens, indirectly contributing to a healthier environment. In low-income countries, there is a focus on using anaerobic digestion technology to improve economic performance and prevent environmental pollution. Here, the utilization of biogas is especially beneficial as it can improve indoor air pollution and provide a sustainable alternative to forest exploitation. The anaerobically produced raw biogas can serve as fuel for cooking and lighting in rural communities, and once refined, it can be utilized for electricity generation. Data from the International Renewable Energy Agency (IRENA) indicates that global biogas capacity for electricity generation grew from 9.5 GW in 2010 to 19 GW in 2019, with electricity output reaching 88 TWh [17]. Although the amount of biogas produced from organic waste varies with conditions, it is estimated that one ton of such waste can yield between 250 and 600 m^3 of biogas. Additionally, a 2018 report estimates that digesting one ton of food waste could generate about 1200 kWh of biogas energy [18].

Chowdhury investigated the anaerobic digestion of different types of solid waste. The results demonstrated an annual methane yield of 1,090,800 m^3 from 40,000 tons of waste, equivalent to producing about 300 m^3 of biogas from one ton of mixed solid waste. This amount generates around 4824 MWh/year of electricity through combined heat power [19]. A study conducted by Sabar et al. on the anaerobic digestion of organic household waste reported a yield of 0.44 m^3 CH_4/Kg within a short retention time of 5 days [20]. Another study on the anaerobic digestion of food waste found that 1550 mL of biogas was generated from 1 g of mixed food waste by day 21 [21]. The energy content of raw biogas is estimated to be around 21.5 MJ/m^3, with 1 m^3 of biogas roughly equivalent to 0.5–0.6 L of diesel fuel or about 6 kWh of electricity. However, in smaller-scale biogas plants, conversion efficiencies as low as 1.7 kWh/m^3 have been reported [22]. This inefficiency may be linked to factors such as suboptimal plant installation, the type of feedstock used, or other operational challenges. In the other case, biomethane has a significantly higher energy content of approximately 36 MJ/m^3. This higher energy density, combined with its potential for net-zero emissions, underscores the importance of investing in biogas plant infrastructure and improving biomethane production through technological advancements. Nowadays, many industries are incentivized by renewable energy benefits to use anaerobic digesters to efficiently manage industrial waste. Given these multifaceted benefits, there is a clear need for further global investigation into the potentials of anaerobic digestion technology.

Waste-to-Resource Approach: Circular Bioeconomy

This century's goal is to switch to ecologically friendly, renewable energy sources in place of nonrenewable ones. Waste-to-resource is a waste management strategy that aims to recover and add value to waste streams. This strategy is closely related to the circular economy, which aims to reduce waste through resource regeneration and recycling. Circular economy is associated with the transition of linear economy into the circular path which introduces novel processes for the efficient utilization and reuse of products and materials, often emphasizing the sale of services. The linear economy begins with raw materials and ends with waste, whereas in the circular economy, the process ends with reusing of the byproducts as a raw material. Practicing the circular economy approach can lead to waste-free value chains and the use of renewable energy and natural resources [23]. Thus, anaerobic digestion is the earliest and best-known technology that approaches achieving a circular bioeconomy. The use of industrial leftovers for sustainable biogas production is in line with the principles of the circular bioeconomy, which minimize feedstock and pretreatment costs. Thus, the conversion of biodiesel feedstocks into biodiesel and then the reusing of the waste as a raw material in the anaerobic digestion process achieves a circular bioeconomy. Figure 2 depicts the bio-circular process of biodiesel by product valorization.

A circular bioeconomy is centered on the replacement of conventional fuels with renewable, biobased alternatives and the reuse of byproducts through the application of biotechnological innovations [24]. The implementation of the bioeconomy approach helps to achieve the sustainable development goals. A sustainable bioeconomy requires law-carbon and low-cost energy input, sustainable supply chain, and the continuous conversion of renewable bioresources into high-value energy. Thus, waste-to-resource technologies are established to which helps to transform low-value biomass into valuable outputs. Hence, a closed-loop circular economy reduces new input materials and waste outputs and achieves the trend of waste into resources [25]. For the sustainable energy production, there is a worldwide requirement to embrace anaerobic digestion technologies. These can utilize a broad array of environmentally friendly substrates with high energy density, including waste biomass from various industries. Organic waste biomass from industrial processes has been considered a viable substrate for biogas production. Hence, anaerobic digestion technology has the potential to aid in the decarbonization of the economy and augment the value of the digestate.

The anaerobic digestion of ethanol byproducts, as investigated by Tena et al. [26], demonstrated an estimated methane production of 3.8 million m^3 per year, with estimated electricity and thermal generation of 14.6 GWh per year and 1.37×10^5 GJ per year, respectively. The replacement of fossil fuel-based energy with this amount of biogas energy would be expected to mitigate up to 7659 tons of CO_2 equivalent per year. On the other hand, if the biogas from the anaerobic digestion process is not managed effectively, the emission of methane to the atmosphere is hazardous, as the global warming potential of methane is 23 times of CO_2 [27]. Hence, balancing the relationship between the natural environment and economic development is very imperative. Thus, advancing in anaerobic digestion technologies is highly demanding for the valorization of biodiesel byproducts that have the advantages of enhancing methane yield, soil quality amendment, protecting against environmental pollution, and job creation.

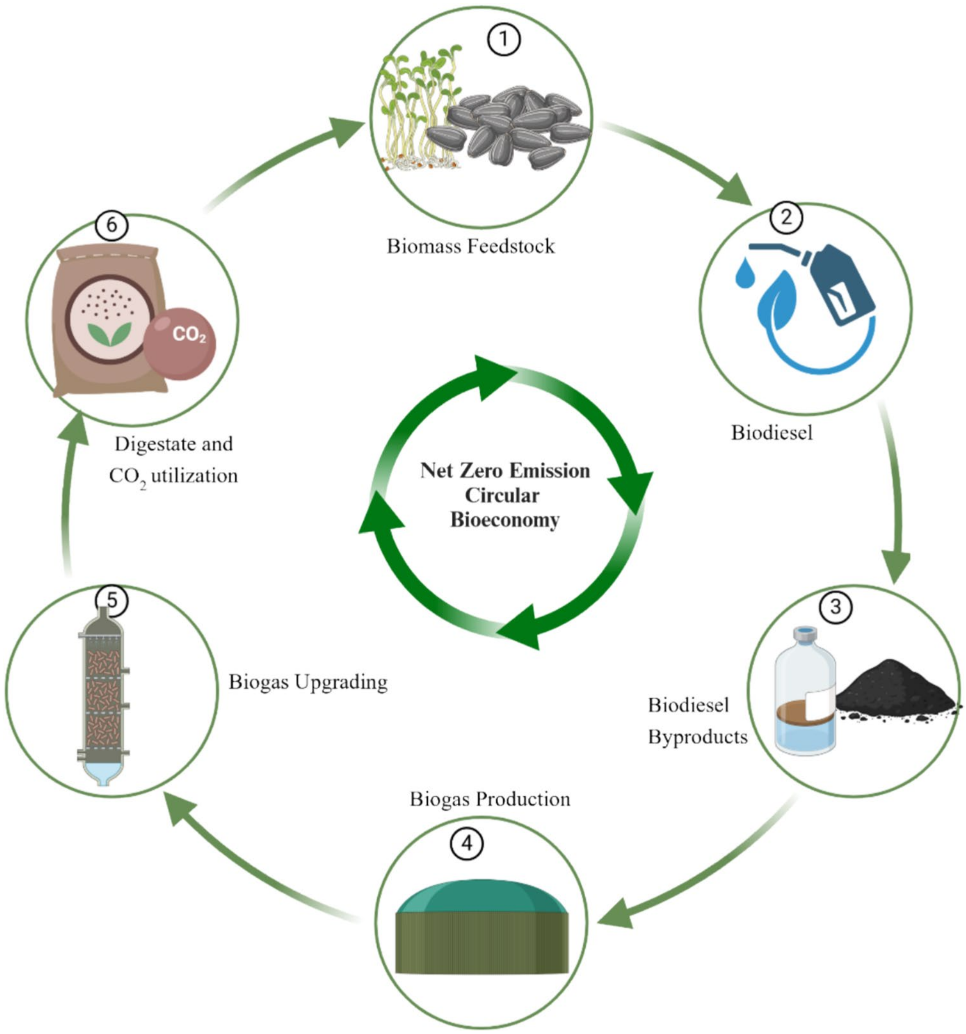

Fig. 2 Biodiesel byproduct valorization: circular bioeconomy (created in https://BioRender.com)

Anaerobic Digestion and Its Financial Return: Techno-economic Analysis

The idea of a circular bioeconomy emphasizes the recovery of all goods from resources without any waste being discharged. In the industrialized world, biogas plants may pay money back in different ways. Techno-economic analysis of anaerobic digestion provides critical information on the economic and environmental feasibility of a biogas plant [28]. Economic viability is an indicator of the overall effectiveness of a project's financial success, while financial viability specifically refers to the profitability of the biogas plant from utilizing biogas and its byproducts. These include the use of a combined heat and power unit to produce electricity and hot water, which further heats the digester. The use of biogas in modified gas boilers also produces hot water for onsite use or export. Moreover, biomethane can be injected into the gas grid for vehicle fuel or used for heat and electricity generation.

Cost–benefit analysis of using biogas as a vehicle fuel is cited as a leading technological indicator of its emerging applications. However, Bentiveglio et al. [29] mentioned that the cost of biomethane production is not yet competitive with the price of natural gas. This is further supported by the data obtained on the economic assessment of evaluating the profitability of upgrading a biogas plant with the substrates of 70% maize ensilage, 10% chicken manure, and 20% vegetable waste from frozen processing. The result highlighted that profitability is verified under certain scenarios; thus, the investment becomes profitable only if changes in the digester diet occur. Additionally, the conversion of the plant is economically not feasible unless subsidies are provided [29]. Clearly, the cost of biomethane production can vary, contingent on the upgrading system implemented and the type of substrate used in the digester. However, the subsidy for upgrading biogas is highlighted as an essential technique to encourage the production of biogas energy from waste, apply the circular approach such as material and nutrient reuse, reduce waste generation, and produce biofuel energy and other value-added chemicals [30].

The profitability of a biogas plant investment can come from cost savings due to the replacement of other energy sources with biogas, income generated from the sale of biogas, time saved when using biogas for cooking or other services, and the use of bio-fertilizer. Financial viability is not always clearly defined, especially for household-based small-scale biogas plants, and some households are hesitant to invest in biogas plant [31]. Therefore, to maximize the utilization of biogas technology, the returns from the biogas plant need to be evaluated and clearly defined. An economic evaluation of a biogas plant, as examined by Bedan et al. [32], and a comparative analysis of costs and benefits through partial budgeting revealed that a small-size biogas plant shows the highest profitability, succeeded by a large-size biogas plant. On the other hand, when considering a discounted cost–benefit analysis, a medium-sized biogas plant proved to be the most advantageous investment, followed by a small-sized biogas plant.

The techno-economic analysis of biogas plants indicates that when biogas is used as a cooking fuel at the household level, replacing liquefied petroleum gas, the estimated payback period is between 5 and 10 years [33]. For instance, the techno-economic analysis of anaerobic digestion of different solid wastes demonstrated a payback period of 5.4 years [19]. This finding suggests that the anaerobic digestion power plant is economically feasible. In the techno-economic analysis and feasibility study of smallholder farm-based anaerobic digestion installation, the author revealed that the installation of 10-m^3 smallholder farm-based digesters is likely to be financially and economically viable; the minimum methane flow of the plant should be 2.5 Nm^3 per day, with 60% methane content which is equivalent to the household energy demand by an average sized low-income south Africa household for cooking with biogas. This is without considering the incomes coming from upgrading the anaerobic digestion systems, digestate utilization, and subsidies brought to society due to the reduction in greenhouse gas emissions [34]. In the other case, the implementation of a smallholder anaerobic digestion plant is financially non-feasible when the daily methane flow is less than 1.53 Nm^3.

In another study, a small-scale household-based digester performing co-digestion of food waste and cow dung found an OLR of 2 kgVS/m^3/day, giving the maximum biogas yield of 4 L/day for a digester volume of 35 L. Based on the OLR, biogas yield, and the volume of the digester, a digester having a volume of above 5 m^3 will fulfill the requirements of three and above family members. An OLR of 2 kgVS/m^3/day will tend to produce a digester of about 5.25 L/day that would be sufficient for the household. The analysis was performed for the digester replacing the liquified petroleum gas both subsidize and non-subsidize, in the household application. The analysis demonstrated that the replacement of LPG with biogas showed a positive net present value with a short payback period [35].

Techno-economic assessment of a small scale of combined heat power (CHP) system on the anaerobic digestion of microalgae utilization from European temperate zones was investigated by Dave et al. [36]. The selected plant size is a community-based anaerobic digestion plant of 1.6 MW microalgae feed rate of 8.64 tons per day. The economic feasibility and detailed mass and energy balance analysis of the process in a combined heat and power plant generated 237 KW of electricity and 367 KW of heat. The electricity selling price is estimated at around € 120/MWh and this is estimated without governmental incentives. The author noted that more systematic studies on cultivation, supply chains, and biogas conversion rates are needed to reduce financial risks.

A techno-economic analysis was conducted on a biogas power plant utilizing palm oil mill effluent as a source of biogas for electricity generation. The oil palm plant, with a capacity of 30 tons of fresh fruit bunches per hour, produced biogas containing 60% CH_4. The project required an investment of 1.979 million USD, with a payback period of 6.6 years, indicating that the plant's construction is feasible. However, it was noted that the project is economically less attractive for investment [37]. Although anaerobic digestion is an efficient technology, it needs to take attention on the capital investment cost and valorization of digestate. Moreover, the circular bioeconomy approach of valorization of anaerobic digestion of biodiesel byproducts could be supported by a numerical data. Thus, the utilization of anaerobic digestion of biodiesel byproducts needs a techno-economic analysis study.

Maximizing Energy Conversion in Anaerobic Digestion

The use of agricultural and industrial wastes for anaerobic digestion is one of the technical interventions of biogas plants [3]. However, the C/N ratio of the feedstocks limits the use of all the industrial and agricultural waste for biogas production. Thus, the C/N ratio is the most important parameter, which shows the potential and biodegradability of the substrates for biogas production. Moreover, the complexity of the digestion process due to the diverse groups of microorganisms in the process and the strict balance between each digestion stage limits the amount of biogas to be generated [30]. Consequently, biogas technology has been extensively studied to improve the conversion process and biogas yield. Thus, the anaerobic digestibility and biomethane production potential of the feedstocks could be enhanced by employing pretreatment techniques, whereas nutrient balance can be maintained by co-digestion of substrates with predetermined chemical compositions. Currently, research has been investigated on the implementation of supplement materials like additives in anaerobic digestion reactors, which further enhances the biogas production potential discussed in the "Additives in Anaerobic Digestion" section.

Maximizing Biogas Yield Through Pretreatment

Pretreatment is an important parameter that highly influences the biogas production potential of the substrates. Some substrates may contain chemicals that inhibit microbial activities and affect the process as well as the biogas yields. To overcome those challenges, various pretreatment techniques, such as physical, biological, chemical, and combined processes, have been developed and applied. The main goal of any pretreatment is to alter or remove any structural and compositional impediments to increase the yields of intended products. The types of pretreatment techniques of anaerobic digestion are depicted in Fig. 3.

The choice of pretreatment depends on the type of feedstock and its composition. Although each method can be applied based on preselection criteria, each technique comes with its own limitations. For instance, physical pretreatment has the advantage of increasing surface area and reducing power requirements. However, it has the disadvantages of not being able to eliminate lignin and rendering the cellulose content inaccessible [38]. Mechanical pretreatment involves size reduction, which is achieved by employing ultrasonic or high-pressure homogenizers. This increases the surface area of the biomass and reduces cellulose crystallinity. The limitations of this method include high energy consumption and the requirement of expensive equipment [39].

On the other hand, chemical pretreatment has the advantage of enhancing the biodegradation of complex materials. The limitations of this method include the risk of chemical hazards, lignin destruction, potential pH increases, and salt build-up [40]. In physicochemical pretreatment, steam explosion decreases biomass crystallinity and removes lignin. This method is effective for hardwood and agricultural residues and can accommodate large-size biomass. However, this method can lead to the formation of inhibitory products like furan derivatives and weak acids, and it needs to be performed at high temperatures and pressures [41]. Contrarily, liquid hot water treatment is a low-cost technique where the biomass is inserted into hot water at high temperatures. This increases the surface area and pore volume of the substrate. The risks associated with this method include sugar degradation and solid loading restrictions to 20% [42].

In biological pretreatment, substrates are processed using microorganisms like bacteria and fungi, as well as enzymes, which offer environmental advantages [43]. However, this approach has limitations, as it is naturally slow and challenging to scale up effectively [44]. Ideal pretreatment methods should be straightforward, affordable, and environmentally friendly and should not produce inhibitory byproducts. Environmentally favorable and time-efficient pretreatments include methods such as hot compressed water, liquid hot water, steam, and superheated steam [45]. Research, however, indicates that no single pretreatment method works universally across all feedstock types. Using a combination of techniques can enhance biodegradation efficiency. For example, combining chemical and thermal pretreatments reduces chemical usage while improving overall pretreatment effectiveness [46]. Examples of increased biogas yields from various pretreatment methods are presented in Table 1.

Studies have reported that combined pretreatments are efficient pretreatment techniques. For instance, Banu et al. [53] investigated waste-activated sludge pretreated by bacterial pretreatment-induced hydrogen peroxide. The result demonstrated that a maximum biomethane yield of 0.174 L/g COD was obtained as compared to the control's only 0.02175 L/g COD and the bacterial pretreatments 0.078 L/g COD. Furthermore, both pretreatment and co-digestion of biogas feedstocks, at the same time better enhance the biogas yield. A study conducted by Zhang et al. [54] investigated the effect of pretreatment and co-digestion of food waste and

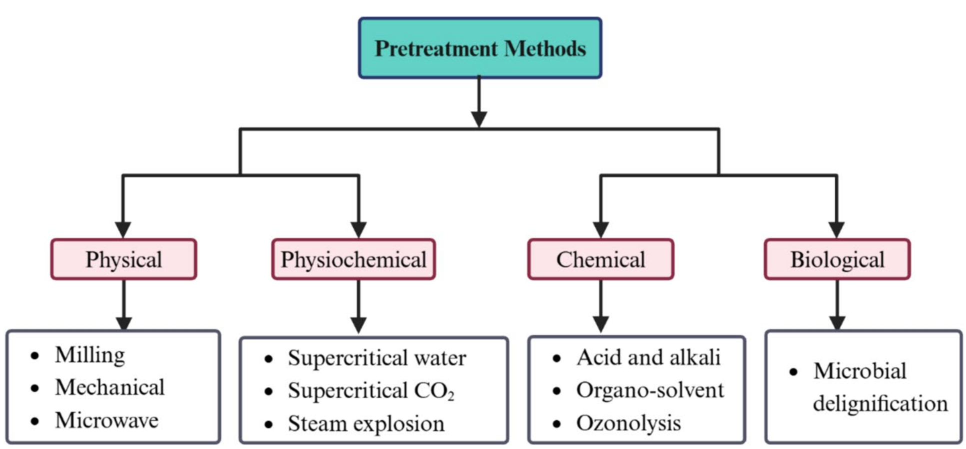

Fig. 3 Biomass pretreatment methods (created in https://BioRender.com)

Table 1 Biogas yield improvement through different pretreatment methods

Feedstocks	Pretreatment employed	Biogas yield	Methane yield enhanced (%)	References
Sewage sludge	Ultrasonic	224 ± 20 mL/g VS	14	[47]
Sewage sludge	Electrochemical (electrooxidation with boron-doped diamond)	312 NL/kg VS	65	[48]
Palm oil mill effluent-activated sludge	Electrochemical oxidation	11.47 mL CH_4/g COD added	94	[49]
Pulp and paper bio sludge	Enzymatic		26	[50]
Miscanthus grass	Alkaline	307 mL g^{-1} VS	74	[51]
Sewage sludge	Microwave		20	[52]

activated sludge on the methane production potential. The experimental result showed that the methane yield obtained from the co-digestion of co-pretreated food waste and waste-activated sludge was around 24% higher than that of the control set of substrates. Similarly, Serrano et al. [55] evaluated the bio-methanation of thermally pretreated sewage sludge, followed by co-digestion with strawberry extrudate. They noted that the combination of thermal pretreatment and co-digestion is effective both in methane production and stability.

There are several studies that explore the possibility of using pretreatment techniques to boost the biogas production potential of biodiesel byproducts. One such study by Jabloniski et al. [56] examined the impact of pretreatment methods on jatropha seed cake for biogas production. They applied thermal and acidic pretreatment to deactivate harmful and antinutritional substances like protease inhibitors. The results demonstrated that thermal pretreatment improved digestion kinetics, decreased protease inhibitor activity, and reduced the concentration of phytate. Additionally, the biogas production potential of jatropha seed cake was benchmarked against that of rape (*Brassica napus*) and flax (*Linum usitatissimum*) seed cakes. About 70% conversion efficiency was achieved from rape and flax seed cakes, whereas jatropha seed cake only achieved about 45% conversion. This lower efficiency could be attributed to the high lignin content in the jatropha seed cake.

Paulista et al. [57] investigated the application of biological and ultrasound pretreatment on the anaerobic co-digestion of sewage sludge and crude glycerol. The experiment was carried out in a mixed batch reactor containing crude glycerol (0.2%, 1.7%, and 3.2% V/V) and sludge, along with the necessary nutrients: potassium nitrate (C:N of 25:1), disodium phosphate (C:P of 120:1), and sodium bicarbonate (2 g/L). According to the findings, the biologically treated samples that underwent anaerobic digestion with 1.7% *Aspergillus niger* and ultrasound treatment at 0.2% crude glycerol (v/v) produced good methane yield. In comparison to the control group, the methane production increased by 11% and 99% for the ultrasound and *Aspergillus niger* biological treatments, respectively. Similarly, Li and Shimazu [58] examined the effect of lipase addition and hydrothermal pretreatment on the co-digestion of food waste and crude glycerol. They noted that a combined pretreatment approach involving both the addition of lipase and hydrothermal pretreatment resulted in an enhanced methane yield.

Maximize Biogas Yield Through Co-digestion

Anaerobic co-digestion is one method of waste treatment in which different wastes are mixed and digested together. Although some biogas feedstocks have the potential to produce methane through anaerobic digestion alone, co-digestion of potential multiple substrates provides opportunities for adequate waste treatment, improving the biogas yield of solid waste and stabilizing bio-slurry. Hence, co-digestion could have complementary benefits such as nutrient balance, toxicity dilution, bulk density, and robust microbes, leading to stable digestion and increasing the methane yield at the same time [59]. Shahbaz et al. [60] studied co-digestion experiments of solid wastes (paper, cardboard, and tissue waste) with food waste and evaluated the impact of the C/N ratio and organic loading rates on their biogas production and digester stability. The result described that the highest biogas production and digester stability were obtained in the co-digestion of paper waste with food waste due to the slow formation of volatile fatty acids, resulting in stable conversion. They demonstrated that the anaerobic co-digestion of the solid wastes (paper, cardboard, and tissue waste) with food waste was an effective conversion, which improved methane yield.

The study of Rahman et al. [61] on co-digestion of kitchen waste and poultry manure under mesophilic conditions also demonstrated increased biogas yield and methane composition, which ensures nutrient balance, buffering capacity, and digester stability. Sarker [62] utilized fish ensilage and fish oil refinery byproducts (glycerol, soap stock, and methyl monoesters) to produce biogas through anaerobic co-digestion under mesophilic conditions. Results showed improved biogas and methane yield was obtained from the co-digestion of the byproducts with fish ensilage compared to fish ensilage alone. Similarly, Housagul et al. [63] investigated on anaerobic co-digestion of banana peel with glycerol, and a maximum biogas yield of 652 mL/g was obtained. Furthermore, several successful studies have also been published on the co-digestion of crude glycerol with sewage sludge [64–67], swine manure [68], dairy wastewater and food wastes [59, 69], fish ensilage [62], and canned seafood [70]. They realized that enhanced biogas yield was observed upon co-digestion of the substrates with crude glycerol compared to the yield of the substrates alone. Moreover, the performance of the anaerobic digestion process is mainly influenced by substrate characteristics (C/N ratio of the feedstock) and digester design.

Maintaining C/N Ratio

The carbon-to-nitrogen ratio is a highly important parameter in the anaerobic digestion process to ensure digester stability and biogas production. Biogas production needs substrates composed of carbon and nitrogen, in which the C/N ratio of the feedstock has a significant effect on the yield. The process demands carbon as the main source of energy for microorganisms, whereas nitrogen is for microbial growth and the synthesis of amino acids and proteins [71]. In reduced form, nitrogen neutralizes the volatile acids produced by fermenting bacteria and favors a pH suitable for microbial growth. Thus, unbalanced nutrients in the substrate highly affect the anaerobic digestion process through the accumulation of ammoniacal nitrogen and subsequent inhibition of microbial activity. The digestion of substrates having a low C/N ratio experiences ammonia inhibition, resulting in a low biogas yield. The study conducted by Choi et al. [72] on the effect of carbon type and C/N ratio on biogas yield revealed that methane production was affected by carbon content, whereas methane production rate is affected by both carbon content and C/N ratio.

Studies have reported that the optimal C/N ratio for anaerobic digestion of organic wastes is 20–35:1. In cases of a high C/N ratio, the system is dominated by acidogenic bacteria, which rapidly consume nitrogen compared to methanogenic bacteria, which leads to methane yield degradation [72]. However, in the actual experiments, the C/N ratio of the substrates in the digester might be too high or too low. Tanimu et al. [73] studied the effect of the C/N ratio of food waste on methane production. The experiment was carried out under mesophilic conditions by mixing different food wastes before anaerobic digestion. The experimental result demonstrated that the highest methane yield was found to be 0.679 L/g VS at the highest C/N ratio of 30:1. They suggested that co-digestion of food waste increases the C/N ratio of the substrate, resulting in pH stability and enhanced methanogenic activity. Glycerol waste from the biodiesel industry primarily consists of carbohydrates with a high C/N ratio [70]. The high C/N ratio of glycerol has a significant advantage in maintaining the C/N ratio of other substrates with a low C/N ratio, like activated sludge, and enhances the biogas yield.

Valorization of Biodiesel Byproducts

Anaerobic Digestion of Biodiesel Byproducts

Biodiesel for transportation is an environmentally friendly and promising alternative energy source due to its renewability and reduction of environmental pollution. Its production is based on a transesterification process that produces fatty acid methyl ester, glycerol, and other minor byproducts. Biodiesel can be used for transport alone or by blending with petroleum in different compositions. However, the high production cost of biodiesel hinders the replacement of conventional fuels with biodiesel. The major byproducts in biodiesel production are (1) crude glycerol generated during the transesterification reaction and (2) seed cake obtained during oil extraction. Hence, the utilization of crude glycerol and other byproducts from the biodiesel industry can minimize the production cost of biodiesel.

Biodiesel byproducts can be recycled through various methods such as anaerobic digestion, composting, biochar, and purification methods. Furthermore, waste glycerol can be managed by incineration, but the carbon released into the atmosphere fails to utilize incineration [74]. Although the purification of crude glycerol is an alternative method, the purification cost makes it an ineffective technique. The conversion of biodiesel byproducts into other usable forms has been extensively investigated, but the conversion techniques are not utilized and are extensively commercialized yet. The symbolic block diagram of the utilization of byproducts from the biodiesel industry is depicted in Fig. 4.

The waste-to-resource conversion process of the biodiesel leftovers through anaerobic co-digestion is summarized in the block diagram (Fig. 4). This block diagram shows the biological degradation of the waste from the biodiesel industry, which is consistent with a closed-loop circular economy mainly focused on the utilization process of leftovers (seed cake from oil extraction and glycerol from transesterification). Moreover, the potential of the digestate produced from anaerobic co-digestion of the leftovers was assessed as a soil conditioner or organic fertilizer. Thus, efficient conversion of the biodiesel leftovers contributes to the environment by producing biogas energy and increasing plant growth and productivity by employing stable digestate for agriculture.

Various studies have been investigated on the production of biogas energy and chemicals from residues and byproducts of the biodiesel industry, as detailed in the

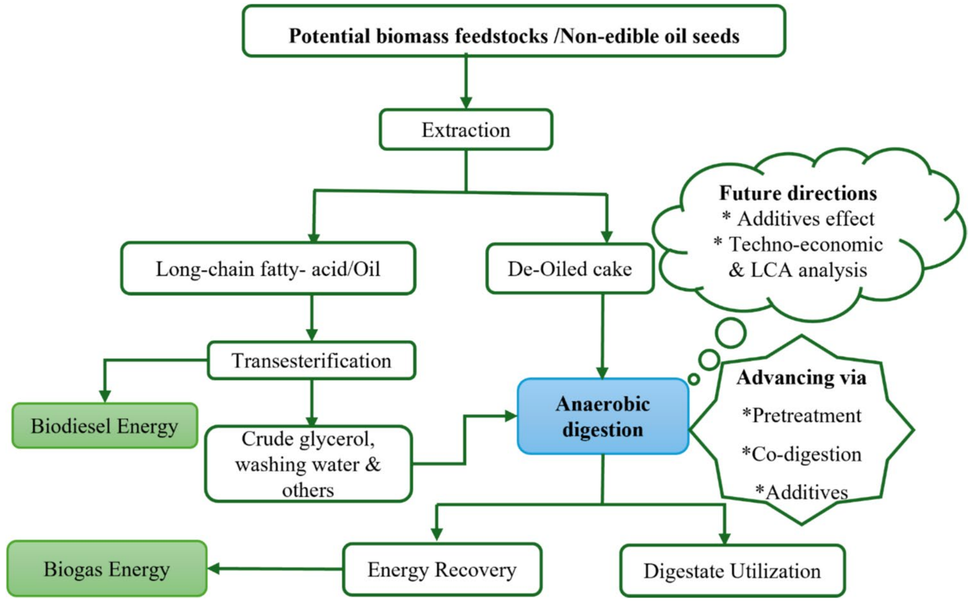

Fig. 4 General description of energy incentive biodiesel industry byproduct valorization process

following section ("Energy Potential and Nutrient Recovery of Biodiesel Byproducts"). Additionally, some review papers have delved into the valorization of biodiesel byproducts. For instance, the first review paper Kolesárová et al. [75] conducted a comprehensive review of biodiesel byproducts such as crude glycerol, washing wastewater, and edible seed cakes for biogas production. Their analysis revealed that these byproducts serve as valuable feedstocks for biogas generation. Consequently, the author suggests the utilization of biodiesel byproducts as an advantageous strategy to reduce the costs of biodiesel production and enhance its competitiveness.

In another study, Jersson Plácido and Sergio Capareda [76] analyzed the transformation of residues and byproducts from the biodiesel industry into products suitable for biorefining. The authors emphasized the potential of these techniques to produce chemicals, bioproducts, and renewable energy from biodiesel waste materials. They also highlighted the environmental benefits and cost reduction potential associated with implementing such conversion methods. Furthermore, they suggested that advancing novel technologies to manage biodiesel industry waste could enhance the sustainability and competitiveness of biodiesel production compared to fossil fuels. Recently, Monthay et al. [77] conducted a review on the biogas production potential of oil seed cakes generated from oil extraction industries, considering edible and non-edible seed cakes. They illustrated that the rich nutritional composition of both the oil cakes makes them viable substrates for biogas generation. Nevertheless, the biogas production potential of these substrates is influenced by factors such as sources of inoculum, co-digestion, and reactor design. The authors also suggested the need for future research to improve the viability of biogas production from oil seed cakes.

In the other case, the present review is mainly focused on the assessment of anaerobic digestion of biodiesel industry residues and byproducts (crude glycerol, non-edible seed cake, and biodiesel washing wastewater). Moreover, the potential of anaerobic digestion techniques such as co-digestion, pretreatments, and supplementing with additives like trace metals and biochar to boost methane yield has been assessed. The most significant experimental results on anaerobic co-digestion, pretreatment, and novel technology utilization were summarized for researchers and other responsible bodies to have easy access to this information. Furthermore, this review assessed the agricultural application of the digestate produced from the anaerobic digestion of biodiesel byproducts, which none of the authors assessed.

Energy Potential and Nutrient Recovery of Biodiesel Byproducts

Anaerobic Digestion of Seed Cake

Additionally, the transition to renewable energy also increases the production of biodiesel from non-edible oily seeds and other potential resources. However, non-edible oil cakes, which have good nutrient content, are left with no applications. Hence, the waste from the industry needs to be explored, which calls for researchers and policymakers to be engaged in the efficient recovery of waste resources. In the biodiesel industry and oil extraction, a huge amount of byproduct known as de-oiled cake or seed cake is produced, which can be obtained from edible seed crops or non-edible seed crops. For instance, Singh et al. [78] noted that the oil extracted from mustard seeds leaves approximately 65% of seed crop residue left over. Additionally, according to the Food and Agricultural Organization (FAO)’s food outlook report, the global production of oil cakes was approximately 158 million tons globally in 2018–2021 [79]. Therefore, in the future, it is anticipated that the amount of seed cakes generated by the industries will rise.

Most of the edible seed cakes could be used as animal feed; however, the toxic nature of the non-edible oil seed cakes does not allow them to be used as animal feed. Many non-edible seed crops can produce oils, such as Jatropha (*Jatropha curcas*), Karanja (*Pongamia pinnata*), Mahua (*Madhuca indica*), silk cotton tree (*Ceiba pentandra*), and castor (*Ricinus communis*). Dumping those seed cakes in the open environment might affect the agricultural sites due to the presence of some contaminants like phorbol esters in jatropha, chromeno flavones in karanja, ricin in castor, and a strong odor of neem [77]. The chemical compositions of non-edible seed cakes are presented in Table 2.

The percentage composition of the seed cakes in Table 2 shows that most of the seed cakes have a higher carbohydrate and crude protein content, which could have the potential for biogas production. Some studies have investigated the potential of seed cakes as feedstock for biogas production such as mahua (*madhuca indica*) and Hingan (*Balanites aegyaptiaca*) [87], date seed [25], jatropha [88, 89], and Pongamia [90] through anaerobic digestion. Hence, the residual fats and oils in seed cakes from the biodiesel byproducts have the potential to generate a significant amount of methane. This section reviews existing research on biogas production from non-edible seed cakes processed through anaerobic digestion or co-digestion methods.

Jatropha seed cake, co-digested anaerobically with varying amounts of paddy straw (used to maintain the C/N ratio), was explored [88]. Various treatments, including cow dung alone, jatropha seed cake alone, and a combination of cow dung and jatropha seed cake, were carried out in batch-type digesters with a hydraulic retention time of 40 days. The maximum gas production was achieved with 20% total solids of the seed cake and a C/N ratio within the range of 22–27:1. The biogas yield per kilogram of total solids of the substrates was compared, and the co-digestion of jatropha seed cake and cow dung was found to have a higher biogas yield than the individual treatments. Similarly, Chandra and colleagues [61] studied methane production from jatropha and Pongamia oil cakes in a floating drum biogas plant under mesophilic condition over 30 days. They found both cakes had high volatile solid content. Jatropha seed cake yielded an average of 0.640 m^3/kg VS biogas and 0.422 m^3/kg VS methane, with a 60% removal efficiency. Pongamia seed cake produced 0.738 m^3/kg VS biogas and 0.448 m^3/kg VS methane, with a 75% removal efficiency. Methane content from jatropha and Pongamia was 66.6% and 62.5% respectively. They also revealed that the biogas produced from the seed cakes contains 15–20% more methane than the biogas produced from cattle dung.

Barik and Murugan [91] also investigated the assessment of the biogas production potential of Karanja (*Pongamia pinnata*) seed cake, a waste from the biodiesel industry. Karanja seed cake was mixed with different percentages of cow dung at a mixed ratio of 75:25, 50:50, 25:75, and 0:100 w/w, and rice straw was also added to maintain the C/N ratio. They evaluated incubation parameters such as pH, temperature, hydraulic retention time, and C/N ratio. They noted that the maximum yield was obtained at the mixing ratio of 25% Karanja to 75% cow dung, whereas the methane and carbon dioxide yields were found to be 73% and 17% respectively. They also revealed that the sample had a slurry of good soil conditioner that was nontoxic and environmentally friendly. Gupta et al. [92] examined the biogas yield of raw and detoxified mahua seed cake. The experiment was performed by applying different treatments that contained raw seed cake, water-treated seed cake, and cow dung. Results showed the maximum biogas yield was obtained from the treatment, which contained 50% detoxified cake and 50% cow dung. They also noticed that anaerobic digestion of hot water-treated mahua seed cake was feasible in terms of biogas yield and the nutritional quality of the digestate.

Gavilanes et al. [93] examined the response of jatropha seed cake to microbial inoculum for biogas and methane production evaluation. The treatments were performed with five types of inoculums (1 control, 3 from jatropha, and 1 from digested swine manure) in a jatropha seed cake stored for 6 and 18 months. Results showed that a maximum methane concentration of around 84% was obtained from the seed cake stored for 18 months and inoculated with the bacterial consortium of isolates from the seed cake. Sharma et al. [2] evaluated biogas production potential from co-digestion of jatropha seed cake and cattle dung in a floating drum digester over 60 days. They used a 1:1 ratio of seed cake to cattle manure with 20% slurry and 80% water. Results

Table 2 Some non-edible seed cake composition

Seed cake	Composition (%)				
	Crude protein	Carbohydrate	Fiber	Ash	References
Jatropha	23–38	32.54	8.11	8.4	[80]
Mahua	30	42.8	8.6	6	[81]
Karanja	13.4	25.2	-	4	[82]
Castor	39.3	24.88	11.6	11.9	[83]
Rubber	36.6	35.35	-	5.5	[84]
Fodder radish	40.19	31.94	5.62	5.01	[85]
Copra	17.8	47.7	10.3	4.9	[86]

showed average biogas production of 0.216 m^3/kg and 0.287 m^3/kg total solid,and 0.252 m^3/kg and 0.335 m^3/kg VS for jatropha seed cake and cattle dung respectively. Methane content was higher in the co-digestion than in cattle dung alone, indicating jatropha seed cake's potential for biogas production due to its volatile solids content.

Ourradi et al. [59] explored methane production from date seed cake in mesophilic conditions using batch-fed mode in a laboratory-scale continuous flow stirred reactor. They bio-stimulated the inoculum with glucose, sodium acetate, and lactic acid. Results showed a 35.93% conversion of organic matter into biogas, with maximum methane production at 3.5 g of VS/L load. A methane yield coefficient of 173.01 NmL CH_4/g of VS was obtained, with the modified Gompertz model providing the best fit for kinetic parameters. They suggested date seed cake could be a viable source for biogas production and fertilizer, recommending further studies on its biodegradability and methane yield.

Anaerobic Co-digestion of Crude Glycerol

Crude glycerol is the primary byproduct generated in biodiesel production, which involves transforming biomass into biodiesel through catalyzed or uncatalyzed transesterification. The process involves the reaction between triglyceride and alcohol, resulting in fatty acid methyl ester (biodiesel) and crude glycerol. The quality and quantity of crude glycerol generated in the biodiesel industry can vary depending on the composition of the feedstock and catalysts used. In the biodiesel industry, crude glycerol accounts for 10% of the total biodiesel produced, which means that for every 100 kg of biodiesel produced, 10 kg of crude glycerol is generated [94]. Rivero et al. [64] state that glycerol from base-catalyzed transesterification typically consists of around 50–60% glycerol, 12–16% alkali soaps and hydroxides, 15–18% methyl esters, 8–12% methanol, and 2–3% water. Nonetheless, the use of heterogeneous catalysts could potentially enhance the quality and quantity of crude glycerol produced in the biodiesel industry.

Documented literature indicates that crude glycerol has found applications in different industries, including direct combustion, anaerobic digestion, and chemical conversion [95]. Moreover, glycerol has been tried to be utilized as animal feed; however, it needs to be effectively purified before use as animal feed [95]. Karlson et al. [96] studied the effect of replacing cow feed with glycerol mixed with grass silage and barley on methane emissions and milk production. They found out that dry matter intake and methane emissions were higher. Direct combustion of crude glycerol is also one way of energy recovery that can substitute for fossil fuel; however, this process is affected by high viscosity, high auto-ignition temperature, and the possibility of forming toxic acrolein [97]. Chemically, crude glycerol can be converted into useful products such as 1,3-propanediol, 1,2-propanediol, dihydroxyacetones, hydrogen, polyglycerols, succinic acid, and polyesters [69].

Biologically, crude glycerol can be used in fermentation or anaerobic digestion for energy recovery. Glycerol is a readily digestible organic substance that can also be easily stored for a long time and is reported as an ideal co-substrate for fermentation and anaerobic digestion [66]. In the biogas industry, the value-added application of crude glycerol has been studied and reported as a potential co-substrate in the anaerobic digestion of other biomass wastes for methane production [67]. However, the problem related to the utilization of crude glycerol for anaerobic digestion is the inhibition of microorganisms associated with the addition of high rates of glycerol. This could be solved by method optimization, metabolic engineering, and the utilization of effective strains. Literature suggests that the addition of crude glycerol in biogas digesters acts as a carbon (C) source, which enhances the biogas yield as well as low nutrient requirements, energy savings, and generation of nutrient-rich stabilized digestate [64].

A study conducted by Aguilar et al. [68] reported anaerobic co-digestion of crude glycerol with swine manure using response surface methodology (RSM). The experiment was carried out using activated sludge as inoculum at different concentrations of swine manure (5–15 g/L) and crude glycerol (4–10 g/L). The highest biogas generated was 521.5 mL/g of COD_{total} at 4 g/L crude glycerol and 5 g/L of swine manure after 35 days. Similarly, Leandro et al. [98] evaluated the biogas production potential of crude glycerol co-digested with swine manure at different concentrations. They suggested that swine manure co-digested with up to 5% (v/v) crude glycerol could be applied as a supplement, which enhances biogas production, methane content, and the removal of total solids, while the addition of a higher concentration of crude glycerol exhibits inhibitory activity.

Crude glycerol from soybean oil and animal fat transesterification was co-digested with food waste and sewage sludge to produce methane and hydrogen. The experiment, under mesophilic conditions, used a two-stage anaerobic co-digestion system. Results showed the highest methane production with 1% glycerol co-digested with other substrates [79]. The methane production potential of crude glycerol from the biodiesel manufacturing industry was also investigated in a large-scale pilot plant by continuously adding crude glycerol to the biogas plant [99]. The experimental result indicated that methane production increased with increasing the load of crude glycerol, and the highest amount of methane was obtained at the supplement of 30 L of glycerol per 30 m^3 day (1 mL per L per day). The energy balance revealed that an energy output equivalent to 106% of the energy output was achieved. Additionally, some of the findings from anaerobic co-digestion of crude glycerol with sludge and other organic waste substrates are summarized in Table 3.

Moreover, anaerobic co-digestion of sewage sludge and crude glycerol could produce methane and value-added chemicals. Kurahashi et al. [32] explored anaerobic co-digestion of sewage sludge and crude glycerol for methane and value-added chemicals production, such as hydrogen and 1,3-propanediol (1,3-PDO). Methane dominated the system, yielding 51.8–77.4 mL/g VS with a small glycerol addition (approximately 0.158 g/g TSS). With higher glycerol amounts (0.158–0.630 g/g TSS), hydrogen was initially produced before methane. Organic acids from glycerol lowered pH and restored methanogenetic activity, counteracting the initial pH rise caused by alkaline impurities in glycerol. Hydrogen fermentation decreased gradually with increased glycerol (> 0.630 g/g TSS), limiting 1,3-PDO generation, while sewage sludge's water content had no impact.

Furthermore, the influence of organic loading rate and retention time in the anaerobic digestion of crude glycerol and other substrates has been investigated. Rivaro et al. [31] examined this impact by co-digesting sewage sludge and 1% crude glycerol in acidogenic and methanogenic digesters on a lab scale. Results showed methane generation primarily occurred during the methanogenetic stage, with methane composition ranging from 50 to 56% CH_4. Hydraulic retention time did not affect the methanogenic stage. However, the organic loading rate influenced both hydrogen and methane yield. The addition of crude glycerol as a co-substrate enhanced methane and hydrogen production potential, particularly at low organic loading rates. Another study conducted by Timmerman et al. [106] on the biogas production effect of decreasing retention time and co-digestion with glycerol demonstrated that the biogas production almost triples at a retention time of 15.6 days and at an addition of 4% glycerol.

Nartker et al. [67] investigated the anaerobic co-digestion of crude glycerol with sewage sludge in a mesophilic condition. The experiment was performed by the systematic addition of glycerol to the digestate up to 60% of the organic loading rate. However, as the feeding schedule indicated, the digesters stayed balanced until microbial inhibition was observed at 70% of the organic loading rate. The outcome demonstrated that a specific biogas yield boost of up to 280% occurred when the loading rate was between 25 and 60%. Compared to sewage sludge alone, biogas production increased by 82% at a 25% organic loading rate. They also demonstrated that high glycerol loading was feasible by adding glycerol gradually to give the microorganisms time to acclimatize.

Mora et al. [88] investigated the effects of glycerol load and pH shocks on a sulfidogenic up-flow anaerobic sludge blanket (UASB) reactor. They found a COD/S-sulfate ratio of 8.5 g O_2 g^{-1} S-SO_4^{2-} achieved maximum sulfate removal efficiency of 100%. Crude glycerol treatment with sulfate in the UASB reactor at low flow velocity and short retention time was successful. Temporary pH shocks (up to pH = 12) inhibited methanogenic activity, significantly affecting UASB performance. Further research is suggested to optimize biodegradability and study volatile fatty acid production from crude glycerol biodegradation in a sulfidogenic UASB reactor. In general, experimental results indicated crude glycerol from the biodiesel industry is a promising co-digestate, enhancing biogas and methane generation. The summary of enhanced biogas yield achieved by the co-digestion of crude glycerol is given in Table 4.

The amount of glycerol shown in summary Table 4 is the optimal amount, which indicates that the addition of crude glycerol to the digestion of organic waste boosts methane yield. However, methane yield enhancements vary depending on the substrate, digestion conditions, and amount of crude glycerol added. In general, the addition of biodiesel byproducts to a biogas reactor demonstrates enhanced biogas yield and stable digestate. In contrast, the biogas yield may have decreased, or the system may have failed at high crude glycerol concentrations. This might be a result of microorganisms being inhibited by crude glycerol at a high

Table 3 Co-digestion of glycerol with different substrates and their experimental findings

Substrate	Conditions	Main result	References
Food waste	Mesophilic and thermophilic	Improved biogas yield at 35 °C at a 3:1 substrate-to-CG ratio	[59]
Sewage sludge	CG (0.63–3%)	Higher biogas yield was obtained at 0.63% CG	[100]
PO decanter cake	CG (7.5–45 g/L), 2-stage	H_2 in stage 1, and CH_4 improved in stage 2	[101]
Sewage sludge	Mesophilic Pilot scale	Maximum CH_4 yield at 1.1% CG	[102]
Wastewater sludge	Evaluation of reactor	Biogas yield boosts up to 1.35% CG	[103]
Dairy wastewater	CG (2%, 4%, and 8%)	Highest biogas yield was generated at 8% CG	[69]
PO mill effluent	Adding CG and ethanol	CH_4 doubled at 1% CG and 5% ethanol mix	[104]
Sewage sludge	Solid retention time effect	1% CG stabilized at a retention of 20–29 days	[105]

CG crude glycerol, *Conc.* concentrations, *PO* palm oil

Table 4 Methane yield enhancements by adding crude glycerol

Substrate	Conditions	CG added (%)	Methane enhanced (%)	References
CG + MSW	Thermophilic	0.25	48	[105]
CG + sludge	BMP and STP	1 and 3	62 and 167	[107]
(CG + ethanol) and palm oil effluent	Batch and mesophilic	1CG and 5 ethanol	50	[104]
CG + dairy WW	Mesophilic	4	226	[69]
CG + sewage sludge	Mesophilic CSTR	3	3.8–4.7	[66]
CG + cattle dung	Mesophilic	5	18	[108]
CG + food waste	Mesophilic batch	1	32	[59]
CG + jatropha seed cake	Mesophilic batch	2	28.9	[89]
CG + seafood WW	Mesophilic	1	108	[70]
CG + pig manure	Mesophilic	1	125	[109]
CG + Sewage sludge	Mesophilic	2	50	[110]

CG crude glycerol, *BMP* biochemical methane potential, *STP* standard temperature and pressure, *WW* wastewater

concentration. Thus, optimizing reaction conditions, feeding rate, treating the substrate, and minimizing crude glycerol impurities employing cost-effective removal technique would be the best alternative solutions to maximize the yield of biogas by anaerobic co-digestion with crude glycerol.

Biodiesel Washing Wastewater for Biogas

Biodiesel washing wastewater is the largest amount of waste produced from the biodiesel industry which is generated during purification of crude biodiesel to remove contaminants. The wastewater generated specifically from alkaline-catalyzed transesterification is mainly composed of methanol, residual oil, glycerol, and soap [111]. Different refining methods have been employed for biodiesel refineries, such as membrane-based technologies and wet and dry washing. Wet washing is performed by using hot and softened water, while the dry method uses adsorbent materials to remove polar contaminants. Both wet and dry washing could be used on a commercial scale; however, wet washing, especially with hot distilled water, is reported as a better technique that fulfills the requirements of the fuel standard [112]. However, for every 1 L of biodiesel, a minimum of 3 L of wash water is generated with a high composition of glycerol and oil, high pH, COD, and BOD [113]. Thus, the washing wastewater generated is reported as highly polluted and should be treated before discharging into the environment.

Various treatments, including physical, physicochemical, chemical, and electrochemical methods, have been used to reduce pollutants in wastewater from biodiesel washing. Treatment usually begins with pretreatment to lower oil and glycerol levels, followed by chemical oxygen demand (COD) reduction. While acidification and coagulation remove grease and oil effectively, they are less successful at reducing COD a limitation that can be addressed by combining these methods with electrochemical treatment. Chemical pretreatment often uses strong acids like sulfuric acid to cut oil content and neutralize residual alkali catalysts. The most common approach, electrocoagulation, generates flocculating agents in an electric field to reduce contaminants like glycerol, oil, and COD. However, this review focuses on using biodiesel wastewater for biogas production through anaerobic digestion, as it offers the most promising treatment with added incentives. Jabra et al. [113] studied the biogas production potential of sludge (anaerobic sludge from a domestic sewage treatment plant) mixed with washing water as one treatment and other treatments with different percentages of biodiesel byproducts using the biochemical methane potential (BMP) test. The experimental result demonstrated that the highest methane generation potential was obtained from the treatment composed of a 9:1 ratio of sludge and washing water from the biodiesel industry. However, the lowest value resulted from the mixture of sludge, glycerin, and press cake.

In the other case, Phukingngam et al. [111] investigated the performance of an anaerobic baffled reactor for organic removal of wastewater and biogas production simultaneously by anaerobic digestion of biodiesel wastewater from a commercial biodiesel-producing factory. It contains a high amount of methanol since the factory has no methanol recovery system, residual oil, glycerin, or soap. Partial removal of the oil was performed by chemical coagulants. The reactor was then operated at 1.5 kg/m^3 per day of COD and 10 days of HRT. The result demonstrated that the best removal efficiencies were provided, and a biogas production rate of 12 L per day with a methane composition of 64–74% was achieved. Additionally, Anderson evaluated the treatment of biodiesel wastewater by anaerobic digestion by applying the biochemical methane potential (BMP) test. The result demonstrated that 185 NmL CH_4/g of COD was obtained. In the other case, wastewater co-digested with crude glycerol was reported to have the highest BMP of 263.8 NmL CH_4/g COD.

Corea et al. [114] examined the biogas generation potential of wastes generated from biodiesel plants using a biochemical methane potential test (BMP). Sludge that was utilized as an inoculum was co-digested with the wastes from the biodiesel byproducts (press cake, glycerin, washing water, oil sludge, and the combination of the wastes). The result demonstrated that the highest biogas generation was obtained from the co-digestion of sludge and washing wastewater. This might be due to the presence of diluted or pre-hydrolyzed oils in the washing wastewater, which favors biodegradability. Anaerobic treatment of wastewater from the biodiesel industry is a low-cost waste treatment technology that also has the advantage of producing methane, which provides the energy for the biodiesel plant to operate. However, the high concentration of chemicals in the biodiesel washing wastewater might be unfavorable for direct application of the wastewater for anaerobic processes. Hence, the wastewater from the biodiesel industry needs to be pretreated before being used for anaerobic digestion. Studies on the treatment and conversion of biodiesel wastewater should be required to use resources effectively and protect water bodies from contamination with toxic contaminants.

Nutrient Recovery and Agricultural Application of Digestate

Bioconversion of biomass into biogas through anaerobic digestion also produces the nutrient-rich digestate, which has the potential to replace inorganic fertilizers. Biogas digestate constitutes a readily available organic nutrient, which can be applied directly to farmland as an organic fertilizer to enhance the nutrient availability of the soil, plant growth, and productivity. Moreover, the utilization of digestate in agriculture contributes to the reduction of inorganic fertilizers and climate change mitigation [115]. However, the digestate needs to be assessed for its chemical composition before being applied to the farmland to protect it from phytotoxicity [25]. Hence, the high electrical conductance associated with the salinity of the digestate, accumulation of heavy metals in the soil, and absorption by the plant pose health risks [116].

Application of Glycerol and Seed Cake Digestate

Crude glycerol from the biodiesel industry has shown potential as a soil conditioner, either used alone or combined with other fertilizers. It supports plant resilience under stressful conditions by sustaining essential metabolic processes. Additionally, crude glycerol enhances soil organic carbon levels, helping prevent soil nitrogen depletion [74]. However, applying large amounts of crude glycerol alone may reduce the soil's available nitrogen, which can be offset by supplementing with nitrogen fertilizers or urea. Moreover, most of the non-edible seed cakes produced from the biodiesel industry have also been reported as potential soil conditioners due to their N, P, K, and high protein content [117]. Mbewe et al. [118] studied the use of *Jatropha curcas* seed cake as a plant growth stimulant and compared it to the application of inorganic fertilizer alone. The seed cake was applied to maize growth and was revealed as a potential natural plant growth stimulant. It could provide a grain yield comparable to that resulting from the application of full inorganic fertilizer.

Similarly, Olowoake et al. [119] evaluated the effect of compost supplemented with jatropha cake on soil fertility and maize yield in Nigeria. They demonstrated that the application of seed cake-supplemented fertilizer increased soil productivity and resulted in a higher maize yield compared to NPK (15–15-15) inorganic fertilizer. Other studies also deduced that the digestate produced from anaerobic co-digestion of crude glycerol with other substrates and seed cakes is mentioned as a stable organic fertilizer [77, 88, 93, 99]. Therefore, the utilization of crude glycerol and seed cake could benefit the biodiesel producers as well as the anaerobic digestion operators, resulting in stability, improved methane yield, and improved soil fertility. Furthermore, the waste generated during anaerobic digestion approximately contains 85–95% of the raw material feed to the digester. Thus, the sustainability of anaerobic digestion becomes challenging without effectively managing the digestate from the anaerobic digestion process.

From an environmental point of view, the management of discharges from the process is a very critical issue. Furthermore, the effective utilization of the anaerobic digestion process depends on the construction of the biogas plant. Additionally, the location of the waste treatment area has its own impact on the environment and economy as well. Developed countries like Sweden have a long-time experience importing waste from Norway, Germany, Denmark, the Netherlands, and Finland. Technically, the importing of non-hazardous wastes could be viable and considered a value addition. However, long-distance transportation of waste could lead to an increase in the carbon footprint of the waste resource. So, biogas plants constructed near the waste generation and accumulation area have both environmental and financial importance.

Challenges and Limitations in Anaerobic Digestion

Despite the numerous benefits of biogas production through anaerobic digestion, there are still challenges and limitations. Some of the limitations include sensitivity to variable loads and organic shocks, high capital costs, high energy requirements, long startup periods, strict control of operating conditions, process instability, low conversion efficiency, and inhibition effects of toxic compounds [120].

Challenges with Feedstocks

Anaerobic digestion has naturally inherent complications depending on feedstock type and process parameters. Inhibition is the primary factor that hinders the anaerobic digestion and methane production potential of substrates. Additionally, process parameters that could lead to the process slowdown of anaerobic digestion or system failure include volatile fatty acid accumulation, slow mass transfer, C/N imbalance, ammonia inhibition, high concentrations of heavy metals, the accumulation of sulfur derivatives, and the recalcitrance of lignocellulosic residues [121]. Thus, the complex and recalcitrant nature of some feedstocks is one source of anaerobic digestion inhibition. Additionally, variability in the quantity and composition of the feedstocks also limits the technological and economic viability of the conversion technology. The above-mentioned anaerobic digestion complications may lead to slower methanogenesis, a lower yield of biogas, and system failure. Hence, it needs to develop strategies to mitigate the anaerobic digestion slowdown or reactor failure of anaerobic digestion process. Traditionally, thus, complications can be improved by concentration optimization, adding buffering agents to maintain pH, balancing C/N by co-digestion, and pretreatment.

Challenges in the Anaerobic Digestion of Biodiesel Byproducts

In the utilization of crude glycerol for biogas production, methanogenic microorganisms could be inhibited by long-chain fatty acids and other contaminants present in crude glycerol [122]. Moreover, the build-up of volatile fatty acids (VFA), such as propionic acid, has been identified as a significant challenge to the stable, long-term operation of anaerobic digestion systems [123]. An accumulation of VFA can cause a sudden decrease in pH, which can inhibit microorganisms and reduce methane yield. This VFA build-up can also suppress ammonia in the anaerobic digestion system. If excess crude glycerol is added while co-digesting with other substrates, it can lead to an accumulation of VFA and ammonia, potentially resulting in system failure. For instance, a study by Sprafke et al. [124] reported system failure during the anaerobic co-digestion of crude glycerol with biowaste. They observed an accumulation of acids and chlorides, accompanied by a drop in methane yield. Therefore, this inhibition could be due to the accumulation of intermediates such as propionic acid, acetic acid, chlorides, and sulfates.

Similarly, Li and colleagues researched the anaerobic digestion of food waste and crude glycerol at concentrations of 5%, 10%, and 15% v/v. They found that adding up to 10% glycerol enhanced methane yield, while adding 15% crude glycerol had a negative impact on the system. This was primarily due to the accumulation of propionic acid, which acidified the reactor and inhibited methanogenic growth [58]. Hence, it is crucial to develop strategies to manage the microbial suppression caused by contaminants in crude glycerol. One alternative approach to the build-up of volatile fatty acids is to bioengineer the anaerobic digestion system to produce volatile fatty acids alongside the primary product [125], effectively transforming biogas plants into biorefineries. Moreover, when converting seed cakes into biogas, the lignin content in some seed cakes may result in lower substrate biodegradability. However, various treatment methods have been employed to address the low accessibility of digestion [77].

Additionally, the high nitrogen content of the seed cakes also hinders the anaerobic digestion process. This could be improved by employing co-digesting with low-cost substrates that can act as carbon source, such as crude glycerol. Moreover, the higher protein, long-chain fatty acids, and lipid composition of the seed cakes might inhibit the microbial activity of anaerobic digestion. This would be improved by maintaining the operating parameters at steady-state conditions. Ewine et al. [89] studied the process optimization of methane yield enhancement in jatropha seed cake and employed alkaline and co-digestion pretreatments. Anaerobic co-digestion of jatropha seed cake was conducted in batch experiments at different feedstock concentrations with varied amounts of crude glycerol (1, 2, 3, and 4%). The experimental result demonstrated that the pretreatment of the jatropha seed cake with 7.32% NaOH and an incubation temperature of 35.86 °C for 54.05 h were reported as the optimum conditions. They noticed that methane yield was significantly increased by alkaline pretreatment and co-digestion with 2% crude glycerol. Furthermore, supplementing supportive materials with the anaerobic digestion reactor further improves the stability of the process and provides an economically viable solution to the anaerobic digestion technology.

Recent Advances in Anaerobic Digestion Technologies

Novel Technologies in the Anaerobic Digestion Process

The primary constraint of conventional anaerobic digestion, leading to decreased methane yield, primarily stems from inhibition. The utilization of additives in anaerobic digestion potentially mitigates the impact of inhibitors and enhances process stability, thereby elevating methane yield [121]. These include active filling, dissolution of metals, zeolites, adsorption, absorption, and alkaline substances [126]. To handle more complex substrates, direct interspecies electron transfer (DIET) through conductive materials is reported as promising anaerobic digestion. Direct interspecies electron transfer is a syntrophic metabolic exchange where electrons flow freely between cells, facilitating interactions without hindrance. Moreover, the methane production potential of the direct interspecies electron transfer (DIET) process with conductive materials is more effective than that of the system with electron carriers [127]. The conductive materials can be biochar, nanotubes, granular carbon, carbon cloth, hematite, and magnetite. Thus, supplementation of conductive materials during anaerobic digestion has shown significant chemical oxygen demand removal, less volatile fatty acid accumulation, and high methane production [120].

Additives in Anaerobic Digestion

Biogas technology is involved in maximizing the biogas yield by utilizing potential feedstocks, adding co-substrate, and applying novel upgrading technologies. Various methods have been implemented in the anaerobic digestion process to enhance the biodegradability of feedstocks and maximize the biogas yield. This includes optimizing substrate concentration, maintaining pH by adding a buffer, co-digesting to balance the C/N ratio or increase organic loading rate, and pretreating to increase biodegradability or enhance the methanogenic stage. Moreover, the addition of some chemical and enzymatic additives and supportive materials is of current research interest in anaerobic digestion to maximize the biogas yield of different substrates. The addition of supportive material includes the addition of a small amount of trace metals, microorganisms, surface-active materials, high-electric conductivity materials, and nanoparticles. Table 5 summarizes some of the anaerobic digestion technology applied additives.

In general, the importance of additives in anaerobic digestors is to support microbial growth, adsorption of inhibitory products, and nutrient supplementation and enhance buffering capacity. This significantly improves biogas generation by enhancing direct interspecies electron transfer (DIET) between methanogens and bacteria. Hence, the use of additives in the anaerobic digestion process is considered an innovative technology; however, some additives have limitations, such as reducing microbial diversity. Depending on the type of substrate and the overall operational conditions, the application of additives like biochar and activated carbon can decrease microbial diversity and elevate pH levels, which may adversely affect the buffering capacity of the anaerobic digestion system. Additionally, the use of additives in anaerobic digestion systems is associated with challenges such as inhibitory effects, difficulties in monitoring, cost inefficiency, and scalability issues [129, 131, 134]. The type of main substrate, the quantity and type of supplements added, and the implementation process significantly influence the efficiency of the anaerobic digestion system. Therefore, it is crucial to optimize the use of additives based on the characteristics of the main substrate and the specific operational conditions. Furthermore, it was suggested that the quality and safety of digestate from anaerobic digestion with additives like activated carbon need further research [135]. The effects of additives on the anaerobic digestion process and methane yield are summarized in Table 6.

The results in Table 6 demonstrate that the addition of biochar and other additives to an aerobic digester enhances methane yield, ranging from 23% to as high as 165%. The highest methane boost was reported with the addition of a small amount of carbon nanotubes. In general, methane yield enhancement primarily depends on the digestion conditions, the nature of the substrate, and the type of additive used. Recently, research interest in the effectiveness of additives in anaerobic digestion systems for various substrates has increased. However, the utilization of biochar and other additives in the anaerobic digestion systems of biodiesel byproducts remains very limited, indicating a significant gap in this area.

Table 5 Some of the additives applied in the anaerobic digestion process and their importance

Additives	Purpose	References
Mineral additives	Act as buffering agent and enhance methane	[128]
Biochar	Good porosity and renewability	[129]
Activated carbon	Can adsorb H_2S from biogas and methane yield enhanced	[130]
Granular activate carbon	Improves DIET and ammonium adsorption Reduces toxic compound mass transfer	[131]
Graphene	Good conductivity and large surface area	[132]
Zeolite	Mitigate ammonia inhibition and improve process stability	[133]
Biological additives	Increase methanogenic and hydrolytic activity Higher waste solubilization	[134]

Table 6 Effects of additives supplemented in anaerobic digester

Substrate	Additives	Findings	References
Cow manure	Mushroom biochar	Methane yield enhanced by 81.3%	[136]
Swine manure	Almond shell biochar	Methane yield increased by 39%	[137]
Water hyacinth	Oak biochar	Biochar < 0.5% boosts CH_4 production	[138]
Chicken manure	GAC	Accelerated biomethane production	[139]
Food waste	(Fe, Co, Ni, Zn, Mn, Cu, Se, and Mo)	CH_4 yield enhanced by 23.8–39.2%	[140]
Animal dung	Fe_2O_3 and TiO_2 nanoparticles	CH_4 yield rise by 119% with 500 mg/L TiO_2 and 121% with 100 mg/L Fe_2O_3	[141]
Poultry waste	Ni nanoparticle	CH_4 rises by 38.4%, upon addition of 12 mg/L of Ni	[142]
Sheep manure	Fullerenes	500 mg/Kg fullerenes enhanced CH_4 yield by 33.6%	[143]
Oleic acid	CNT	Adding of 1 g/L CNT, CH_4 yield increased by165%	[144]

GAC granular activated carbon, *CNT* carbon nanotubes

Additives for Biodiesel Byproducts

While research on biogas production from biodiesel byproducts has been conducted, there is a need for improved utilization of waste from the biodiesel industry through advancements in anaerobic digestion technology. Enhancing biogas yield can be achieved by carrying out anaerobic digestion of substrates with additives such as iron. Guiza et al. [134] reported that iron (in the forms of Fe^0 and Fe^{3+}) shows promising results, acting as both an electron pair donor and acceptor, and serving as a cofactor in enzymatic processes. Sen et al. [145] investigated the application of additives, co-digesting jatropha seed cake and bagasse with iron (Fe^{2+}) as a value additive. They found that co-digesting seed cake with bagasse at a 2:1 ratio resulted in the highest biogas yield of 0.136 m^3/Kg VS, compared to jatropha seed cake alone, which only produced 0.064 m^3/kg VS. When 10 mM Fe^{2+} was added to the co-digestion of seed cake with bagasse, biogas yield improved to 0.203 m^3/kg VS. This suggests that co-digesting jatropha seed cake with bagasse, along with an optimized amount of Fe^{2+}, could boost biogas yield within a shorter digestion time.

In the other case, Schmidt et al. [146] studied the anaerobic digestion of jatropha curcus seed cake and the effect of iron additives on biogas gas quality and process stability during increasing organic loading rate (OLR). The increase in OLR of 1.3 to 3.2 g_{vs} L^{-1} d^{-1} without the addition of an iron additive decreases the methane concentration from 69.3 to 44.4%. However, the decline is less pronounced in the presence of iron additive; this is due to the removal of H_2S, which protects the inhibition of the process. They highlighted that the addition of iron additives in the anaerobic digestion of jatropha seed cake improves process stability and biogas quality. Bocaro et al. [147] investigated the effects of Fe_2O_3 nanoparticles in the anaerobic co-digestion of crude glycerol and sewage sludge. Experimentally determined that the reactor supplemented with 0.2 gL^{-1} of Fe_2O_3 nanoparticles enhanced the methane production by 49.8%. They mentioned that the Fe_2O_3 nanoparticle optimizes the bacterial and archaeal communities of the anaerobic digestion process.

Additionally, Watanabe et al. [148] investigated the effect of the use of cider charcoal as a support material for microorganisms in the co-digestion of crude glycerol and sewage sludge. They found that the methane composition of the charcoal-containing reactor is 1.6 times higher than the amount of methane in the controlled reactor. Moreover, the concentration of propionate, which is the main disrupting compound in glycerol, was lower in the charcoal-supplied reactor compared to the controlled one. In another study, Im et al. [149] investigated methods to improve the efficiency of anaerobic digestion of glycerol by promoting direct interspecies electron transfer reactions (DIET) with magnetite. The addition of magnetite to the reactor up to 7.5 g COD/L resulted in a 16% increase in methane yield. Recently, Li and Shimizu [123] studied the effect of biochar addition and re-inoculation on the anaerobic co-digestion of crude glycerol and food waste. They demonstrated that significantly enhanced methane yield and process stability were observed in the reactors with added biochar compared to the control and re-inoculation reactors. However, the advancement of biogas production technologies for the byproducts of the biodiesel industry is not utilized. Future research is needed to enhance the biogas production potential of the byproducts using novel technologies by employing low-cost and potentially supportive materials.

Future Perspectives and Recommendations

A sustainable bioeconomy emphasizes the transition from conventional fuels to renewable, biobased resources, driven by innovative conversion technologies. A key strategy for advancing a circular bioeconomy is the sustainable conversion of renewable bioresources into net-zero-emission fuels. This approach ensures that atmospheric carbon released during biobased energy use is reabsorbed through photosynthesis, effectively balancing emissions and achieving a net-zero impact. Anaerobic digestion technology, for instance, transforms a variety of biodegradable wastes into biogas, which positively impacts public health, the environment, and the economy. Furthermore, the circular bioeconomy focuses on the optimal conversion of bioresources and the utilization of wastes from one industrial process as inputs for another, thereby enhancing the efficiency of conversion technologies. Consequently, integrating biodiesel and biogas production significantly contributes to the circular bioeconomy and a cleaner environment. However, if the byproducts of biodiesel production are not effectively utilized, the economic benefits of biodiesel production remain questionable. Studies have identified seed cake and crude glycerol as promising options for biogas generation. For optimal biogas yield, adjusting the C/N ratio is desirable, which can be enhanced by co-digesting low-ratio substrates with high-ratio substrates.

Most of the co-digestion processes using crude glycerol and seed cake discussed in this review showed positive results, suggesting that co-digestion improves biogas and methane yield. Additionally, to effectively utilize biodiesel byproducts for biogas production, biogas plant sites should be located close to biodiesel facilities. Approximately 80% of the original substrate remains in the digester as digestate, which is rich in nitrogen and phosphorus. To apply the digestate directly to fields, it would be ideal to construct biogas plants near agricultural areas. This approach could be economically viable, as it reduces production and transportation costs. Although the anaerobic digestion of biodiesel byproduct is a promising technology, its profitability and techno-economic feasibility are not yet clearly defined. Therefore, several factors must be addressed to improve cost-effectiveness, maximize yield, ensure profitability, and enhance the sustainability of biodiesel waste conversion. Thus, further research should focus on the following:

(1) Exploring complementary feedstocks for co-digestion with biodiesel byproducts, focusing on available and easily biodegradable options that simultaneously boost methane yield potential.
(2) Conducting detailed techno-economic analyses to evaluate the cost-effectiveness of biodiesel byproduct valorization. Thus, the adoption of innovative technologies depends on their economic feasibility and their ability to ensure environmental safety.
(3) Advancing anaerobic digestion with innovative technologies, including additives and other supportive materials. Although improved anaerobic digestion through the use of additives has shown enhanced methane yields, more focus is needed to standardize these methods in a sustainable and cost-effective manner.
(4) Assessing the co-digestion potential of seed cakes (high in nitrogen) and crude glycerol (high in carbon) in the presence of additives or supporting materials, which may further enhance their stability, nutrient balance, and buffering capacity, thereby improving biogas generation.
(5) Ensuring the safety, stability, and agricultural usability of digestate produced during the anaerobic digestion process, particularly when additives are used, especially in cases involving high concentrations of heavy metals or other toxic substances.

Conclusion

This review highlights the potential of biodiesel byproducts like glycerol and non-edible seed cakes for biogas production, emphasizing their high nutrient content as promising substrates for anaerobic digestion. It underscores the benefit of using advanced techniques like co-digestion and pretreatment to boost methane production and convert waste into bioenergy. Despite challenges related to feedstock nature, crude glycerol impurities, and lignin content, these can be mitigated by balancing chemical composition, optimizing loading rate, and using additives. Furthermore, the resulting digestate can serve as organic fertilizer. The review advocates for advanced anaerobic digestion technologies and interdisciplinary research to ensure sustainability and scalability.

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Data Availability All the articles, books, and reports used in this review paper are published and available.

Declarations

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